

STT 861: Theory of Probability & Statistics I

Lecture 5: Conditional Probability

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Reference: Casella & Berger Ch. 1.3

Lecture 5 Outline

Today

- Conditional probability – definition & intuition
- Multiplication rule
- Law of Total Probability
- Card and medical examples

Conditional Probability

Definition

For events A, B with $P(B) > 0$, the **conditional probability** of A given B is:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Intuition: Restrict sample space to B ; re-normalise.

Two Cards – Diamonds

Draw 2 cards without replacement. $A = \{1\text{st is } \diamond\}$, $B = \{2\text{nd is } \diamond\}$.

$$P(A \cap B) = P(B|A) \cdot P(A) = \frac{12}{51} \cdot \frac{13}{52} = \frac{156}{2652} \approx 0.059$$

More generally, the Multiplication Rule for n events $A_1, \dots, A_n$ says:

$$P(A_1 A_2 \dots A_n) = P(A_1)P(A_2|A_1)P(A_3|A_1 A_2) \dots P(A_n|A_1, \dots A_{n-1})]$$

Multiplication Rule & Total Probability

General Product Rule

$$P(A_1 \cap \cdots \cap A_n) = P(A_1) P(A_2 | A_1) P(A_3 | A_1 A_2) \cdots P(A_n | A_1 \cdots A_{n-1})$$

Law of Total Probability

If $A_1, \dots, A_k$ partition S with $P(A_i) > 0$:

$$P(B) = \sum_{i=1}^k P(B | A_i) P(A_i)$$

Examples

Quality Control

Box: 10 components (7 good, 3 defective). Draw 3 without replacement.

$$P(\text{all good}) = \frac{7}{10} \cdot \frac{6}{9} \cdot \frac{5}{8} = \frac{210}{720} \approx 0.292$$

Card Drawing – Total Probability

Let $A = \{\text{1st card } \diamond\}$, $B = \{\text{2nd card } \diamond\}$.

$$P(B) = P(B|A)P(A) + P(B|A^c)P(A^c) = \frac{12}{51} \cdot \frac{13}{52} + \frac{13}{51} \cdot \frac{39}{52} = \frac{663}{2652} = \frac{13}{52}$$

The probability that the second card is a diamond equals 13/52 – same as the first!

Conditional Probability is a Valid Probability

$P(\cdot|B)$ Satisfies Kolmogorov Axioms

For fixed B with $P(B) > 0$, $Q(A) = P(A|B)$ is a valid probability. Proof:

(A1) $Q(A) = P(AB)/P(B) \geq 0$. (A2) $Q(S) = P(SB)/P(B) = 1$. (A3) If A_i disjoint: $Q(\bigcup A_i) = \frac{P((\bigcup A_i)B)}{P(B)} = \frac{\sum P(A_i B)}{P(B)} = \sum Q(A_i)$. $\square$

Two Aces (CB Ex 1.3.1)

Draw 2 cards from 52. $A = \{1\text{st} = \text{ace}\}$, $B = \{2\text{nd} = \text{ace}\}$.

$$P(B|A) = 3/51. \quad P(A \cap B) = P(B|A)P(A) = \frac{3}{51} \cdot \frac{4}{52} = \frac{1}{221} \approx 0.0045.$$

Law of Total Probability – Proof and Example

Law of Total Probability

If $A_1, \dots, A_n$ partition S and $P(A_i) > 0$:

$$P(B) = \sum_{i=1}^n P(A_i)P(B|A_i)$$

Proof: $B = BS = B(\bigcup A_i) = \bigcup (BA_i)$. Since A_i are disjoint, so are BA_i .
 $P(B) = \sum P(BA_i) = \sum P(A_i)P(B|A_i)$. $\square$

Game Show Revisited (Monty Hall)

You are a contestant in a game show. There are three doors, behind one of which is a car. You pick one of the doors. Then, the game show host shows you that behind one of the other two doors there is no car. You are given the opportunity to change your choice. Should you stay with your door or should you switch?

Let $W_i = \{\text{car behind door } i\}$ ($P = 1/3$ each). You pick door 1. Host reveals goat behind door 3. By total probability: $P(\text{win by switching}) = P(W_2|H_3) = 2/3$. Always switch!

Breast Cancer Screening – Bayes in Action

CB Example: Mammography Screening

Women aged 40–65: prevalence $P(C) = 0.00167$.

Sensitivity: $P(+|C) = 0.90$. Specificity: $P(-|\bar{C}) = 0.99$.

$$\begin{aligned} P(C|+) &= \frac{P(+|C)P(C)}{P(+|C)P(C) + P(+|\bar{C})P(\bar{C})} = \frac{0.90 \times 0.00167}{0.90 \times 0.00167 + 0.01 \times 0.99833} \\ &= \frac{0.001503}{0.001503 + 0.009983} = \frac{0.001503}{0.011486} \approx \mathbf{0.131} \end{aligned}$$

Despite positive test: only $\sim 13\%$ probability of cancer.

If negative: $P(C|-) = P(C|+, \text{neg test}) \approx 0.00017$.

Activity 5 – Conditional Probability

Activity 5

- Ⓐ Machine I makes 60% of output (3% defective); Machine II makes 40% (5% defective). Find $P(\text{item is defective})$.
- Ⓑ Bowl 1: 3R+2W; Bowl 2: 1R+4W. Pick a bowl at random (50/50), then draw a ball. Find $P(\text{Red ball})$.
- Ⓒ **Bonus:** In (a), given that an item is defective, $P(\text{from Machine I})$?

Activity 5 – Solutions

- (a) $P(D) = 0.03 \times 0.6 + 0.05 \times 0.4 = 0.018 + 0.020 = \mathbf{0.038}$
- (b) $P(R) = P(R | B_1) \frac{1}{2} + P(R | B_2) \frac{1}{2} = \frac{3}{5} \cdot \frac{1}{2} + \frac{1}{5} \cdot \frac{1}{2} = \frac{4}{10} = \mathbf{0.4}$
- (c) $P(M_1 | D) = \frac{0.018}{0.038} \approx \mathbf{0.474}$